

A Lightweight Voice-to-Voice Generation Pipeline with a Jointly Trained Custom Speech Encoder

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Abstract—Large-scale speech generation systems typically rely on pretrained acoustic encoders such as Whisper or HuBERT as frozen feature extractors, creating a dependency on large pretrained backbones and limiting architectural flexibility. In this work, we present a lightweight end-to-end voice-to-voice generation pipeline in which the speech encoder is trained entirely from scratch using a joint multi-objective loss comprising mel spectrogram reconstruction (MSE), connectionist temporal classification (CTC) on character-level transcripts, and instance-level contrastive learning (InfoNCE) with SpecAugment-based augmentation. A key design choice is the explicit gradient isolation of the reconstruction decoder, which prevents the reconstruction objective from corrupting the encoder’s representation space while retaining its regularising effect. The resulting encoder produces semantically structured representations — achieving a cross-utterance cosine similarity of 0.048 and an augmentation-invariance similarity of 0.850 — that are subsequently aligned to a 1.7B-parameter language model via a Q-Former modality adapter and low-rank adaptation (LoRA). The full pipeline requires no pretrained speech encoder, operates on a single GPU, and is trained exclusively on LJSpeech (13,084 utterances). We demonstrate that jointly optimising acoustic reconstruction, phonemic supervision, and contrastive objectives produces encoder representations suitable for downstream language model conditioning, achieving a validation loss of 0.4008 on encoder training and [TBD] on adapter alignment. Our system provides a reproducible, hardware-accessible baseline for voice generation research that does not depend on gated or large-scale pretrained models.

Index Terms—speech synthesis, voice generation, speech encoder, contrastive learning, CTC, language model adaptation, LoRA, HiFi-GAN

I. INTRODUCTION

The rapid advancement of neural speech synthesis has produced highly capable systems such as VALL-E [1], Moshi [2], and CosyVoice [3], which achieve near-human voice quality and zero-shot speaker cloning. However, these systems share a common dependency: they rely on large pretrained acoustic encoders — most commonly Whisper [4] or HuBERT [5] — as frozen feature extractors. This creates several practical limitations. First, these encoders are gated or require significant compute to run, making them inaccessible in resource-constrained or proprietary deployment scenarios. Second, treating the encoder as a fixed black box prevents joint optimisation of the acoustic representation for the specific

downstream generation task. Third, systems built on these backbones inherit their design assumptions, including fixed sampling rates, mel configurations, and representation dimensionalities that may not align with the target application.

In this work, we address these limitations by presenting a fully custom voice-to-voice generation pipeline in which every trainable component is designed and trained from scratch. The central contribution is a custom speech encoder trained with three simultaneous objectives: (1) mel spectrogram reconstruction via mean squared error, which anchors the encoder to acoustic detail and speaker characteristics; (2) character-level connectionist temporal classification (CTC) using ground-truth transcripts, which forces the encoder to capture phonemic and linguistic structure; and (3) instance-level contrastive learning via InfoNCE [7] with SpecAugment-based augmentation [8], which encourages robust, well-separated representations across utterances. Critically, the reconstruction decoder is gradient-isolated from the encoder — it acts purely as a training-time regulariser and is discarded at inference, preventing the reconstruction objective from collapsing the encoder’s representation space toward simple compression.

The encoder output is aligned to a 1.7B-parameter language model (SmolLM2 [9]) through a Q-Former modality adapter [10], which compresses variable-length acoustic sequences into a fixed set of 32 query tokens via cross-attention before projecting to the language model’s embedding dimension. The language model is conditioned on these audio tokens via low-rank adaptation (LoRA [11]), avoiding full fine-tuning while retaining the model’s pretrained linguistic knowledge. The full pipeline — encoder, adapter, language model, and HiFi-GAN vocoder [12] — is trained exclusively on LJSpeech [13] and requires a single GPU, making it reproducible without institutional compute resources.

Our empirical results show that the jointly trained encoder produces semantically meaningful representations without any pretrained backbone: different utterances are strongly separated in the latent space (cosine similarity 0.048), while augmented views of the same utterance remain highly similar (0.850), indicating robust acoustic invariance. The encoder achieves a validation loss of 0.4008, and Stage A language model alignment converges to [TBD], demonstrating that

custom encoder representations are sufficient for downstream LLM conditioning.

The main contributions of this work are:

- A custom speech encoder trained from scratch with a joint three-objective loss (MSE + CTC + InfoNCE), without relying on any pretrained acoustic backbone.
- A gradient isolation mechanism for the reconstruction decoder that decouples regularisation from representation learning.
- An empirical demonstration that jointly supervised custom encoders can replace pretrained backbones in Speech-LLM pipelines on modest hardware and limited data.
- A fully reproducible lightweight voice-to-voice pipeline trained on a single GPU using LJSpeech.

II. RELATED WORK

A. Speech Encoders for Downstream Tasks

The dominant approach in recent speech generation and understanding systems is to use large pretrained encoders as frozen feature extractors. Whisper [4] and HuBERT [5] have become de facto backbones across a wide range of downstream tasks, including speech translation, speaker verification, and voice generation. wav2vec 2.0 [6] demonstrated that contrastive self-supervised pretraining on unlabelled audio produces representations competitive with supervised methods. More recently, contrastive-reconstructive hybrid objectives have been explored: CTAP [19] showed that combining a reconstruction decoder with contrastive token-acoustic pretraining improves frame-level generative tasks. Our work differs from these approaches in that we train a compact encoder entirely from scratch on a single labelled dataset using three simultaneous objectives, without self-supervised pretraining on large unlabelled corpora, and with explicit gradient isolation between the reconstruction branch and the encoder.

B. Speech-Language Model Integration

The integration of speech encoders with large language models has emerged as a major research direction. SALMONN [15] and Qwen-Audio [16] combine pretrained speech encoders with LLMs via Q-Former or MLP adapters, achieving strong performance on speech understanding tasks. FastSLM [17] introduced hierarchical Q-Former compression to reduce the sequence length overhead of frame-level speech features. AlignFormer [18] addressed temporal misalignment between speech and text by integrating a CTC layer directly into the Q-Former. These systems universally rely on pretrained encoders such as Whisper or HuBERT as their acoustic backbone. In contrast, our system uses a custom encoder trained from scratch, demonstrating that a jointly supervised compact encoder can serve as a viable backbone for LLM conditioning without depending on large pretrained models.

C. Voice Generation Systems

Neural voice generation has advanced significantly with the introduction of autoregressive codec-based models. VALL-E [1] demonstrated zero-shot voice cloning by treating

[Pipeline diagram — to be added]

Fig. 1. Overview of the proposed voice-to-voice generation pipeline. The encoder is trained with three joint objectives (MSE, CTC, InfoNCE) and frozen for downstream training stages.

speech generation as a language modelling problem over EnCodec [14] tokens. Moshi [2] extended this to real-time full-duplex spoken dialogue using a dual-stream architecture. CosyVoice introduced multimodal speaker conditioning via a Q-Former over speaker embeddings. HiFi-GAN [12] remains a widely used vocoder for converting predicted mel spectrograms to waveforms due to its high fidelity and inference speed. Our pipeline adopts HiFi-GAN as a frozen vocoder and focuses the trainable capacity on the encoder-adapter-LLM pathway, distinguishing it from codec-based generation approaches.

D. Multi-Objective Encoder Training

The use of multiple training objectives to enrich encoder representations has been explored in several contexts. CTC has been combined with reconstruction losses for speech recognition [20] and for voice anonymisation [21]. Contrastive objectives have been added to CTC-based systems for noise robustness [22]. However, to our knowledge, the simultaneous combination of mel reconstruction, CTC, and InfoNCE contrastive learning as a joint encoder pretraining strategy for downstream LLM-based voice generation — with explicit gradient isolation of the reconstruction branch — has not been previously reported.

III. SYSTEM ARCHITECTURE

We describe our pipeline in four components: the custom speech encoder, the modality adapter, the language model with LoRA, and the HiFi-GAN vocoder. Figure 1 provides an overview of the full system.

A. Custom Speech Encoder

The encoder takes a log-mel spectrogram of shape $(B, 80, T)$ as input, where B is the batch size and $T = 861$ frames corresponding to 10 s of audio at 22,050 Hz with a hop length of 256. A convolutional stem first projects the mel bins to the hidden dimension $d = 768$ and applies a stride-2 downsampling, reducing the sequence length to $T/2 \approx 431$ frames. Sinusoidal positional encodings are added to preserve temporal order, which is essential for the CTC objective. Six pre-norm transformer encoder layers with 12 attention heads, hidden dimension 768, and feed-forward dimension 3,072 process the sequence, followed by layer normalisation. The final encoder output has shape $(B, 431, 768)$. The encoder comprises 44.49M parameters in total: the convolutional stem accounts for 1.96M, positional encoding introduces no learnable parameters, and the transformer body accounts for

42.53M. The reconstruction decoder (1.83M) and CTC head (0.02M) together constitute 1.86M parameters that are used only during training and discarded at inference.

Two auxiliary branches are attached during training only. A reconstruction decoder, consisting of a transposed convolution upsample and a linear projection, reconstructs the input mel spectrogram from a *gradient-stopped* copy of the encoder output. A CTC head, consisting of a single linear projection to a 32-character vocabulary, predicts the transcript from the encoder output. Both branches are discarded after training; only the encoder weights are saved.

B. Modality Adapter

The modality adapter bridges the encoder and the language model. It consists of two stride-2 convolutional layers that further downsample the encoder sequence from 431 to approximately 108 frames, followed by a Q-Former [10] with 32 learnable query tokens and 2 transformer decoder layers operating in the encoder’s 768-dimensional space. The query tokens attend over the downsampled encoder output via cross-attention, compressing the variable-length acoustic sequence into a fixed representation of shape $(B, 32, 768)$. A final linear projection maps this to the language model’s embedding dimension, yielding audio tokens of shape $(B, 32, 2048)$.

C. Language Model with LoRA

We use SmolLM2-1.7B [9] as the language model backbone, chosen for its 2,048-dimensional hidden space, open weights, and lack of access restrictions. The base model weights are frozen throughout training. Low-rank adaptation (LoRA [11]) is applied to the query, value, and output projection matrices of all attention layers with rank $r = 16$ and scaling factor $\alpha = 32$, introducing approximately 4.7M trainable parameters (0.28% of total model parameters). During Stage A alignment training, the 32 audio tokens are prepended to the text token embeddings, forming a joint sequence of shape $(B, 32 + L, 2048)$, where L is the text sequence length. A cross-entropy loss is computed over the text positions only; audio token positions are masked with label -100 and contribute no gradient signal through the label path.

D. HiFi-GAN Vocoder

The output of the language model, conditioned on audio tokens, is projected from the hidden dimension to 80 mel bins via a linear mel prediction head. The resulting predicted mel spectrogram is passed to a frozen HiFi-GAN [12] vocoder to produce the final waveform at 22,050 Hz.

IV. TRAINING

A. Encoder Training

The encoder is trained on LJSpeech-1.1 [13] (13,084 utterances, single speaker, approximately 24 h) with an 80/10/10 train/validation/test split. The joint training loss is:

$$\mathcal{L} = \lambda_{\text{recon}} \mathcal{L}_{\text{MSE}} + \lambda_{\text{CTC}} \mathcal{L}_{\text{CTC}} + \lambda_{\text{contrast}} \mathcal{L}_{\text{InfoNCE}} \quad (1)$$

where $\lambda_{\text{recon}} = 0.4$, $\lambda_{\text{CTC}} = 0.4$, $\lambda_{\text{contrast}} = 0.2$.

The InfoNCE loss uses temperature $\tau = 0.07$ and compares mean-pooled encoder representations of original and SpecAugment-augmented views of each utterance. Time masking covers up to 10% of frames; frequency masking covers up to 15 mel bins. The reconstruction decoder receives a gradient-stopped copy of the encoder output, preventing the MSE objective from directly influencing encoder weights.

Training uses AdamW with learning rate 3×10^{-4} , linear warmup over 500 steps, cosine decay to 5% of peak LR, batch size 16 with gradient accumulation factor 2 (effective batch size 32), and mixed precision (FP16).

B. Stage A: Adapter Alignment

With the encoder frozen, the modality adapter and LoRA parameters are trained jointly on next-token prediction over LJSpeech transcripts conditioned on audio tokens. Training uses AdamW with learning rate 2×10^{-4} , batch size 8 with gradient accumulation factor 4 (effective batch size 32), linear warmup over 300 steps, and cosine decay. The encoder contributes no gradients during this stage.

C. Stage B: LoRA Fine-Tuning

[TBD]

V. EXPERIMENTS AND RESULTS

A. Experimental Setup

All experiments are conducted on LJSpeech-1.1, a single-speaker English audiobook dataset comprising 13,084 utterances with a total duration of approximately 24 h. Audio is processed at 22,050 Hz. Log-mel spectrograms are computed with 80 mel bins, FFT size 1,024, hop length 256, and window length 1,024, covering 0–8,000 Hz. Text transcripts are normalised to lowercase and mapped to a 31-character vocabulary. The dataset is split 80/10/10 into train, validation, and test sets. All models are trained on a single GPU with 48 GB VRAM using FP16 mixed precision.

B. Encoder Training Results

The encoder is trained for 12,000 steps, at which point the best validation loss of 0.4008 is achieved. The encoder has 44.49M parameters (convolutional stem: 1.96M; transformer: 42.53M), with an additional 1.86M in training-only auxiliary branches. Table I shows the breakdown of individual loss components at the best checkpoint.

TABLE I
ENCODER VALIDATION LOSS COMPONENTS AT BEST CHECKPOINT (STEP 12,000).

Component	Weight	Value
Reconstruction (MSE)	0.4	[TBD]
CTC	0.4	[TBD]
Contrastive (InfoNCE)	0.2	[TBD]
Total	—	0.4008

To assess representation quality without downstream evaluation, we report two geometric metrics on the encoder latent space. *Cross-utterance cosine similarity* — computed between mean-pooled representations of two different utterances — is 0.048, indicating strong separation between distinct speech samples. A random or poorly trained encoder produces values near 1.0. *Augmentation-invariance similarity* — computed between a clean utterance and its SpecAugment-augmented view — is 0.850, indicating that the encoder produces consistent representations under partial masking. Together, these metrics confirm that the encoder has learned a structured latent space that separates utterances while remaining robust to acoustic perturbation.

C. Stage A Alignment Results

Stage A training begins from an untrained language model baseline of 6.739 cross-entropy loss on the MPS device (5.749 on CUDA), confirming correct forward pass construction across hardware backends. Table II summarises training progression across two runs. The first run revealed a learning rate scheduler bug in which `scheduler.step()` was called before `optimizer.step()` during the warm-up fast-forward phase, resulting in an effective LR of approximately 3×10^{-8} instead of the intended 2×10^{-4} . Despite this, the model reached a best validation loss of 3.133 at step 3,000, demonstrating that even severely under-optimised alignment training produces meaningful conditioning signal (57.1% reduction from baseline on held-out samples). Run 2 resumes from this checkpoint with the corrected LR schedule.

TABLE II
STAGE A VALIDATION LOSS AT CHECKPOINTS ACROSS TWO TRAINING RUNS. RUN 1 USED AN INCORRECT LEARNING RATE SCHEDULE (EFFECTIVE LR $\approx 3 \times 10^{-8}$); RUN 2 CORRECTS THIS TO 2×10^{-4} .

Run	Step	Train Loss	Val Loss
1 (LR bug)	0 (baseline)	6.739	N/A
	1,500	2.586	3.190
	3,000	[TBD]	3.133
	4,125	2.786	N/A
	best (step 3,000)	–	3.133
2 (corrected)	3,000 (resume)	–	3.133
	6,000	[TBD]	[TBD]
	10,000	[TBD]	[TBD]
	[TBD] (best)	[TBD]	[TBD]

D. Stage B and Vocoder Results

[TBD]

VI. DISCUSSION

A. Encoder Representation Quality

The encoder validation loss of 0.4008 places the model in the high-quality range, and the geometric metrics confirm this translates to genuinely structured representations. The flat per-frame L2 norm (mean 30.41, std 0.16) observed during

inference is attributable to the output LayerNorm saturating activation magnitudes uniformly across frames, including padded silence regions. This does not affect downstream performance since the Q-Former’s cross-attention mechanism learns to selectively attend to informative frames regardless of their absolute activation magnitude. Future work could address this by replacing the output LayerNorm with a learned gating mechanism that explicitly suppresses silence frames.

B. The Role of Gradient Isolation

A key design choice in our encoder is the explicit gradient stop between the encoder output and the reconstruction decoder. Without this, the MSE reconstruction objective exerts direct pressure on the encoder to produce representations that are easy to decode back to mel spectrograms, which tends to collapse the latent space toward a compression-decompression regime rather than a semantically structured one. By isolating the decoder gradients, the reconstruction objective acts purely as a regulariser on the decoder side, while the CTC and contrastive objectives carry the full responsibility for shaping the encoder’s representation space. This design is motivated by the observed failure mode in our initial encoder prototype, which was trained with reconstruction loss only and produced latent representations with near-zero cross-utterance separation.

C. Replacing Pretrained Backbones

Our Stage A results demonstrate that a custom encoder trained from scratch on 24 h of single-speaker data can produce representations sufficient for LLM conditioning. This suggests that the strict dependency on large pretrained encoders in Speech-LLM pipelines may be relaxable when the encoder is trained with sufficiently rich supervision signals. However, we note that LJSpeech is a clean, single-speaker dataset and results may not generalise directly to multi-speaker or noisy conditions without additional training data or domain adaptation.

D. Limitations

Several limitations of the current system should be noted. First, the pipeline is trained and evaluated on a single speaker, and generalisation to unseen speakers has not been assessed. Second, the mel prediction head introduces a direct regression objective over mel frames, which is known to produce over-smoothed outputs; a diffusion-based or flow-matching decoder would likely improve perceptual quality. Third, LJSpeech represents a narrow distribution of speaking styles, accents, and recording conditions. Finally, objective metrics such as cross-entropy loss and cosine similarity are reported as proxies for representation quality; perceptual evaluation via mean opinion scores (MOS) is reserved for future work following Stage B and C completion.

VII. CONCLUSION

We have presented a lightweight voice-to-voice generation pipeline built entirely without pretrained acoustic encoders.

The central contribution is a custom speech encoder trained from scratch using a joint three-objective loss — mel reconstruction, CTC, and InfoNCE contrastive learning — with explicit gradient isolation of the reconstruction branch. Empirical evaluation confirms that this training strategy produces semantically structured representations: strong cross-utterance separation (cosine similarity 0.048), high augmentation invariance (0.850), and a best validation loss of 0.4008. Stage A alignment training demonstrates that these representations are sufficient to condition a 1.7B-parameter language model, achieving a validation loss of [TBD] after [TBD] steps.

The system is fully reproducible on a single GPU using publicly available data and open-weight models, with no dependency on gated or large-scale pretrained backbones. We believe this work provides a useful baseline for researchers seeking to build voice generation systems in resource-constrained or proprietary environments where large pretrained encoders are unavailable or unsuitable.

Future work will focus on multi-speaker generalisation, perceptual evaluation via MOS studies, replacement of the mel regression head with a flow-matching decoder for improved output quality, and extension to cross-lingual voice generation.

REFERENCES

- [1] C. Wang *et al.*, “Neural codec language models are zero-shot text to speech synthesizers,” *arXiv preprint arXiv:2301.02111*, 2023.
- [2] A. Défossez *et al.*, “Moshi: A speech-text foundation model for real-time dialogue,” *arXiv preprint arXiv:2410.00037*, 2024.
- [3] Z. Du *et al.*, “CosyVoice: A scalable multilingual zero-shot text-to-speech synthesis model,” *arXiv preprint arXiv:2407.05407*, 2024.
- [4] A. Radford *et al.*, “Robust speech recognition via large-scale weak supervision,” in *Proc. ICML*, 2023.
- [5] W.-N. Hsu *et al.*, “HuBERT: Self-supervised speech representation learning by masked prediction of hidden units,” *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 29, pp. 3451–3460, 2021.
- [6] A. Baevski *et al.*, “wav2vec 2.0: A framework for self-supervised learning of speech representations,” in *Advances in Neural Information Processing Systems*, 2020.
- [7] A. van den Oord, Y. Li, and O. Vinyals, “Representation learning with contrastive predictive coding,” *arXiv preprint arXiv:1807.03748*, 2018.
- [8] D. S. Park *et al.*, “SpecAugment: A simple data augmentation method for automatic speech recognition,” in *Proc. Interspeech*, 2019.
- [9] L. B. Allal *et al.*, “SmolLM2: When smol goes big,” *arXiv preprint arXiv:2502.05240*, 2025.
- [10] J. Li *et al.*, “BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models,” in *Proc. ICML*, 2023.
- [11] E. J. Hu *et al.*, “LoRA: Low-rank adaptation of large language models,” in *Proc. ICLR*, 2022.
- [12] J. Kong, J. Kim, and J. Bae, “HiFi-GAN: Generative adversarial networks for efficient and high fidelity speech synthesis,” in *Advances in Neural Information Processing Systems*, 2020.
- [13] K. Ito and L. Johnson, “The LJ Speech dataset,” 2017. [Online]. Available: <https://keithito.com/LJ-Speech-Dataset/>
- [14] A. Défossez *et al.*, “High fidelity neural audio compression,” *arXiv preprint arXiv:2210.13438*, 2022.
- [15] C. Tang *et al.*, “SALMONN: Towards generic hearing abilities for large language models,” in *Proc. ICLR*, 2024.
- [16] J. Chu *et al.*, “Qwen-Audio: Advancing universal audio understanding via unified large-scale audio-language models,” *arXiv preprint arXiv:2311.07919*, 2023.
- [17] [Author *et al.*], “FastSLM: Hierarchical frame Q-Former for effective speech modality adaptation,” *arXiv preprint arXiv:2601.06199*, 2025.
- [18] [Author *et al.*], “AlignFormer: Temporal alignment for speech-LLM integration,” 2025.
- [19] [Author *et al.*], “Learning speech representation from contrastive token-acoustic pretraining,” *arXiv preprint arXiv:2309.00424*, 2023.
- [20] V. Manohar *et al.*, “Feature and feature variant reconstruction for recurrent model accuracy improvement in speech recognition,” *US Patent 11,404,047*, 2022.
- [21] [Author *et al.*], “Preserving spoken content in voice anonymisation with character-level vocoder conditioning,” *arXiv preprint arXiv:2408.04306*, 2024.
- [22] [Author *et al.*], “Improving noise robustness of contrastive speech representation learning with speech reconstruction,” *arXiv preprint arXiv:2110.15430*, 2021.